

Momentum is All You Need

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Code

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1 Background & Research

Exploration

Our team began by exploring a range of technical indicators to support a momentum-driven trading strategy. Through heuristic experimentation and testing, we found that many did not give us positive Sharpe in the ways we used them. Through this experimentation, we established our current strategy.

1.1 RSI in conjunction with Bollinger Bands

We began by exploring RSI for stock selection, and Bollinger Bands (when broken) for momentum signals¹. However, as we began testing, we noticed that many different combinations of these strategies did not yield us a positive Sharpe ratio. We hypothesize that this is because RSI is generally a weak indicator on its own but even in combination with Bollinger Bands only tells us if a stock is behaving unusually. However, if there is a more longer term trend (e.g. AI boom in the last few years), we do not expect something like NVDA to mean revert back to what it used to be. This is why the strategy we did choose accounted for longer trends as well.

1.2 SMA

We also explored strategies using SMA, and EMA/WMA for momentum signaling². While we did heuristically test using them, we found that on its own, SMA and its alternatives did not tell us much. Instead, we found several examples in literature and elsewhere where it was used effectively in more quantitatively advanced strategies, like LSTM³. However, these strategies are typically very complex and required mathematical applications out-of-scope, so we decided not to proceed with them.

An alternative strategy that we did explore was moving crossover SMAs (double & triple), and this yielded quite positive results. However, this discovery occurred after we had decided on our present strategy, and had begun tuning parameters. As such, we decided to focus on our existing approach.

1.3 Auto-Correlation (Current Strategy)

Our final strategy is a pure, long-only momentum system applied to five pre-selected stocks chosen for high auto-correlation (see next section). The algorithm's logic is based primarily on a daily assessment of volatility and recent momentum. We calculate a custom weighted_volatility measure by averaging the 5-day, 10-day, and 20-day volatility, with greater importance given to the more recent periods.. This use makes our approach conceptually similar to Bollinger Bands, as we use volatility to set dynamic trading thresholds rather than fixed price targets. The trading entry/exit signals are straightforward. We initiate a long position by allocating 40% of the portfolio to a stock if its most recent daily return is positive and

¹ <https://arxiv.org/pdf/1212.4890>

² <https://arxiv.org/html/2504.10914>

³ <https://arxiv.org/html/2509.14401>

exceeds 90% of this weighted volatility. The exit signal is if the stock's daily return turns negative, we immediately liquidate the entire position.

A key point is the lack of shorting in our strategy. We tried several strategies that incorporated short positions but largely found that they didn't improve returns, drawdown, or Sharpe ratio. This was the case with both our current indicators, or modified indicators. We hypothesize that it is likely because of strong upward bias in highly autocorrelated stocks (where momentum tends to persist more on long) and determined that without robust risk controls, shorting only needlessly increases risk.

2. Algorithm

2.1 Finding Stocks

The first step of our strategy involved finding securities with high positive auto-correlation. We built on the innate S&P 500 stock cap by only looking at the top 100 companies ranked by market capitalization. This was done to ensure that the stocks selected were stable fundamentally, decreasing risk. From these, we manually computed 1-day auto-correlation of returns for each stock and sorted. One-day autocorrelation (by this definition) is a simple algorithm that checks the previous day's and today's return, and finds the correlation in data points. Figure 1 depicts an example of one of our used stocks.

One intentional choice we made was to ignore industry diversification. While we did initially select for high auto-correlation in 5 independent industries, with the hypothesis that this would reduce risk through decorrelation and improve Sharpe, this actually yielded inferior results on several past test timeframes. This promoted our conviction in high auto-correlation stocks. Another intentional choice we made was to look for the highest auto-correlation stocks of the year prior to our testing/training period. This is to only use data we'd have available, inferring the stocks would continue with their trend-following behavior. This also prevented any avoid look-ahead bias. Our five highest securities were [HD](#), [ROI](#), [AVGO](#), [BAC](#), and [MS](#).

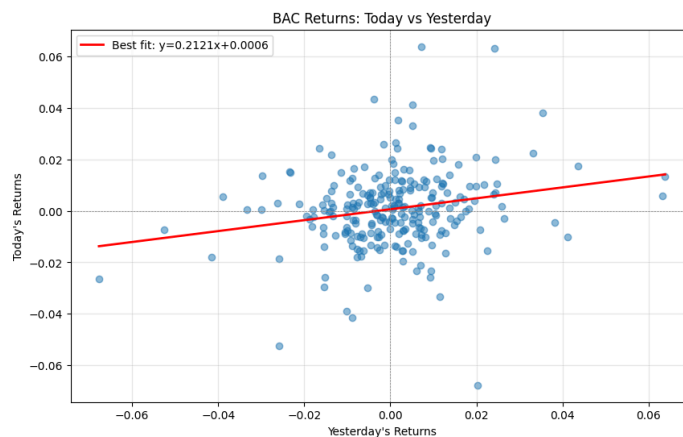


Figure 1 - Auto-correlation of BAC. Value: 0.21

2.2 Volatility

Our entry investment logic follows each security independently. That is, when we enter/liquidate a stock is independent of how we invest in the other stocks. Our entry condition for the 5 securities is detailed in the next passage, but relies primarily on stock volatility. To achieve an accurate lookback volatility, our calculation utilized weighted volatility. This calculates volatility over three different windows – 5, 10, & 20 days – then averages them. We tested this weighting against a standard 20-day volatility measure and found it produced a higher Sharpe ratio, indicating this weighing better accounts for risk signals. We believe this to be the case because recent trends in volatility are more impactful on the returns, and weighted volatility gives more weight to recent returns.

2.3 Entry Condition

Our buy signal triggers when a stock's daily return exceeds 0.9 times its weighted volatility. This indicates that a stock is following a trend and we want to utilize its momentum for higher returns (we utilized long-only, as shorting yielded poor results). Conversely, our sell signal triggers either when the returns are negative or the return is below 0.1 times the weighted volatility – meaning the trend has halted.

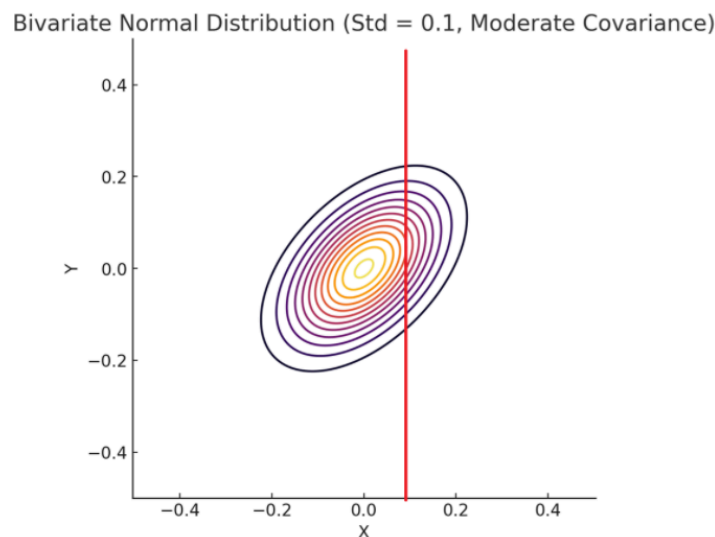


Figure 2 - Entry Condition Applied to Bivariate Distribution

Figure 2 displays why our strategy works in theory. X and Y are random variables each representing returns, and are thus each represented by a normal distribution by GBM. Assume that the combined bivariate distribution has positive correlation (thus positive covariance). Each point (x,y) represents a pair of consecutive returns. The red line represents the entry condition ($0.9 * \text{std}$). When $x > (0.9 * \text{vol})$, we invest, hoping that $y > 0$. Because of the characteristics of normal distributions $Y|x \sim N(x, \sigma^2)$. Thus, $E[y] > 0$. In other words, if today's return is high, $E[\text{tomorrow's return}]$ high as well. While y (tomorrow's return) is above 0, we keep making money.

2.4 Position Sizing and Portfolio Construction

Initially, we set position weights at 0.2 to ensure we could invest even if all 5 stocks gave positive signals. However, we found that we were rarely fully invested. Over the training period, the distribution of invested stocks per day showed 57 days with zero positions, 52 days with one position, 28 days with two positions, 33 days with three positions, 12 days of 4 stocks, and 2 days of 5 stocks. The weighted average of invested stocks at any given time came out to 1.4 stocks held per day. This means on average, 3.6 out of

5 possible positions were uninvested, corresponding to 72% of the portfolio sitting in cash ($3.6 \times 0.2 = 0.72$). Since idle cash does not generate returns, this is unsatisfactory.

We adjusted position sizing to 0.4 per stock, for a max of 2.5 positions. Although we initially thought this would limit us due to missing more price signals, we are able to capitalize more on the stocks we do invest in frequently. The result of this new weight leads us to holding an average of less than 50% cash, lower than the previous 72%. The number 0.4 came through manual testing and follows intuitively, since we statistically rarely invest in more than 2 securities. Note that we can exceed more than 100% of our portfolio in QC automatically via leverage and we recognize that, but since it is a part of the documentation and the system checks for max borrowing amount, we are going to infer that is allowed.

3. Results

3.1 Test & Training Set

Our training set was from 01/07/2024 - 01/03/2025 and the test set was from 01/03/2025 - 01/07/2025. Utilizing this 2:1 split, we avoided look-ahead bias by optimized the weights/boundaries on the training set only. This is how we came up with 0.4 for weights and 0.9 and 0 for entry/exit boundaries. We can see from Table 1 that our drawdown & Sharpe threshold in both periods are being satisfied.

	Return	Sharpe Ratio	Drawdown
Training Set	20.03%	1.157	8%
Testing Set	12.47%	1.198	9.4%

Table 1 - Results from training and test sets

3.2 Out of Performance Set

	Return	Sharpe Ratio	Drawdown
OOP Set	3.7%	0.337	4%

Table 2 - Results on out of performance set

Our selection algorithm returns ["BAC", "CVX", "MMC", "UNH", "C"] on the year prior to the out-of-performance set. Using these stocks we see the results in Table 2. While not as good as our training and test set, the returns are still positive with a decent Sharpe ratio and (extremely) low drawdown.

4. Conclusion

This momentum-based trading strategy exploited positive auto-correlation in SPY stocks, achieving strong risk-adjusted returns with Sharpe ratios exceeding 1.15 and drawdowns below 10% in both training and testing periods. The use of weighted volatility & optimal position sizing (0.4 per stock) enabled

effective trend capture while maintaining capital efficiency. While out-of-performance results were weaker, positive returns and low drawdown suggest the strategy remains viable due to our robust stock selection.